

Transitivity $\Rightarrow$ Quasi-transitivity

Result — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Premises

- **Transitivity.** For all $p, q, r \in \Delta(X)$: if $p \succcurlyeq q$ and $q \succcurlyeq r$ then $p \succcurlyeq r$.

0.2 Conclusion

- **Quasi-transitivity.** Strict preference is transitive: if $p \succ q$ and $q \succ r$ then $p \succ r$.

0.3 Proof

Let $p \succ q \succ r$. Then $p \succcurlyeq q \succcurlyeq r$, so $p \succcurlyeq r$. If also $r \succcurlyeq p$ then $r \succcurlyeq p \succcurlyeq q$ gives $r \succcurlyeq q$, contradicting $q \succ r$. So $p \succ r$.

Record: topics/decision-theory/results/transitivity-implies-quasi-transitivity.yaml